

Digital Image Processing (CS-323)

Lecture-8

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- Topics to Learn

1. Histogram Matching (Specification)
2. Histogram Matching (Specification)-Numerical

Histogram Processing: Histogram Matching

- **Histogram equalization** produces a transformation function that seeks to generate an output image with a **uniform histogram**.
- When **automatic enhancement** is desired, this is a good approach to consider because the results from this **technique are predictable and the method is simple to implement**.
- However, **there are applications in which histogram equalization is not suitable**.
- **In particular, it is useful sometimes to be able to specify the shape of the histogram that we wish the processed image to have.**
- The method used to generate images that have a specified histogram is called **histogram matching** or **histogram specification**.

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Histogram Processing: Histogram Matching

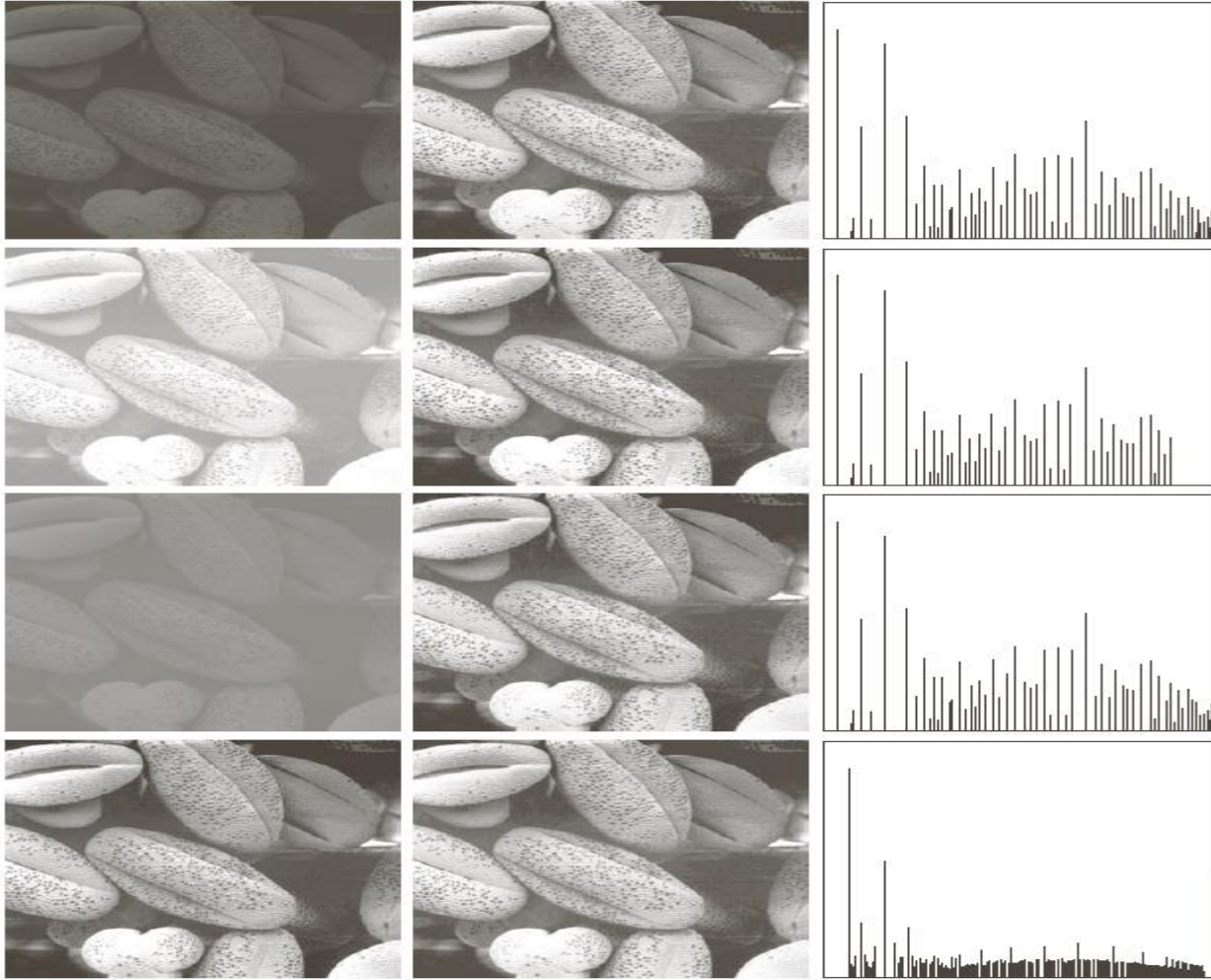


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

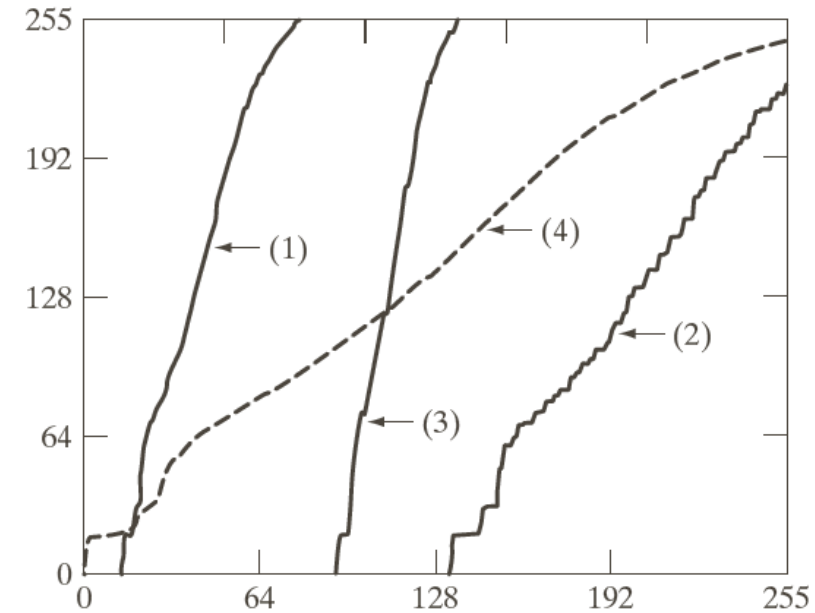


FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

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Image Processing,
Gonzalez 3rd
edition)

Histogram Processing: Histogram Matching

- Consider continuous intensities r and z as random variables with PDFs $p_r(r)$ and $p_z(z)$ are pdf , respectively.
- Here, r and z denote the intensity levels of the input and output (processed) images, respectively.
- We can estimate $p_r(r)$ as the pdf from the given input image, and $p_z(z)$ is the specified pdf that we wish the output image to have.

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Histogram Processing: Histogram Matching

- Let s be a random variable with the property
- $s = T(r) = (L-1) \int_0^r p_r(w) dw \dots\dots\dots(\text{Eq. 1})$

Suppose next we define a random variable z with the property

- $G(z) = (L-1) \int_0^z p_z(t) dt = s \dots\dots\dots(\text{Eq. 2})$
- where t is dummy variable of integration. It then follows from these two equations that $G(z) = T(r)$, and therefore, that z must satisfy the conditions
- $z = G^{-1}[T(r)] = G^{-1}[s]$

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Histogram Processing: Histogram Matching

- Let s be a random variable with the property

$$\begin{aligned}s = T(r) &= (L-1) \int_0^r p_r(w) dw \\ &= (L-1) \int_0^r \frac{2w}{L-1^2} dw \\ &= \frac{2(L-1)}{(L-1)^2} \int_0^r w dw \\ &= \frac{2(L-1)}{(L-1)^2} \frac{r^2}{2}\end{aligned}$$

$$\text{or, } \mathbf{s = T(r) = \frac{r^2}{L-1}}$$

$$\mathbf{G(z) = (L-1) \int_0^z \frac{3w^2}{(L-1)^3} dw = \frac{3(L-1)z^3}{3(L-1)^3}}$$

$$\mathbf{G(z) = \frac{z^3}{(L-1)^2} \Rightarrow z = [(L-1)^2 s]^{1/3}}$$

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- Now

$$\mathbf{s} = \mathbf{T(r)} = \frac{\mathbf{r^2}}{\mathbf{L-1}}$$

$$\text{and } z = [(L - 1)^2 s]^{1/3}$$

$$\text{Therefore, } \mathbf{z} = \left[(L - 1)^2 \frac{\mathbf{r^2}}{\mathbf{L-1}} \right]^{1/3} = \mathbf{[(L - 1)r^2]^{1/3}}$$

- Thus, the pixel intensity value z of the histogram specified image can be directly generated by above equation.

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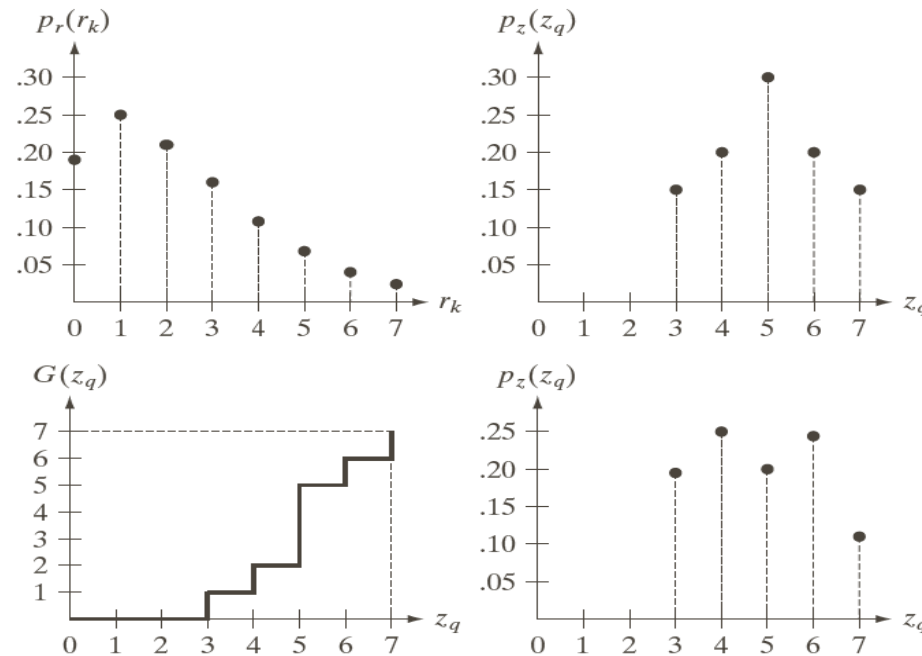
Histogram Processing : Histogram Matching

- Consider 64x64 hypothetical image in Table 3.1. It is desired to transform this histogram so that it will have the values specified in the second column table 3.2.

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11

TABLE 3.2
Specified and actual histograms (the values in the third column are from the computations performed in the body of Example 3.8).

(Courtesy: Digital Image Processing, Gonzalez 3rd edition)



a b
c d

FIGURE 3.22
(a) Histogram of a 3-bit image. (b) Specified histogram. (c) Transformation function obtained from the specified histogram. (d) Result of performing histogram specification. Compare (b) and (d).

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Histogram Processing : Histogram Matching

- Solution: The histogram- equalized values are given below.

$s_0 = 1.33 \rightarrow 1$	$s_1 = 3.08 \rightarrow 3$	$s_2 = 4.55 \rightarrow 5$	$s_3 = 5.67 \rightarrow 6$
$s_4 = 6.23 \rightarrow 6$	$s_5 = 6.65 \rightarrow 7$	$s_6 = 6.68 \rightarrow 7$	$s_7 = 7.00 \rightarrow 7$

- In the next step, all the values of transformation function:
- $G(z_0) = (L-1)\sum_{j=0}^0 p_z(r_j) = 0.00$
- $G(z_1) = (L-1)\sum_{j=0}^1 p_z(r_j) = (8-1)[p_z(z_0) + p_z(z_1)] = 0.00$
- $G(z_2) = (L-1)\sum_{j=0}^2 p_z(r_j) = (8-1)[p_z(z_0) + p_z(z_1) + p_z(z_2)] = 0.00$
- $G(z_3) = 1.05, G(z_4) = 2.45, G(z_5) = 4.55,$
- $G(z_6) = 5.95, G(z_7) = 7.00$

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Histogram Processing : Histogram Matching

$G(z_0) = 0.00 \rightarrow 0$	$G(z_1) = 0.00 \rightarrow 0$	$G(z_2) = 0.00 \rightarrow 0$	$G(z_3) = 1.05 \rightarrow 1$
$G(z_4) = 2.45 \rightarrow 2$	$G(z_5) = 4.55 \rightarrow 5$	$G(z_6) = 5.95 \rightarrow 6$	$G(z_7) = 7.00 \rightarrow 7$

- For example $G(z_3) = 1.05 \rightarrow 1$, this implies that intensity value of $s = 1$ (obtained after histogram- equalization) will transform to 3.

z_q	$G(z_q)$
$z_0 = 0$	0
$z_1 = 1$	0
$z_2 = 2$	0
$z_3 = 3$	1
$z_4 = 4$	2
$z_5 = 5$	5
$z_6 = 6$	6
$z_7 = 7$	7

TABLE 3.3

All possible values of the transformation function G scaled, rounded, and ordered with respect to z .

s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7

TABLE 3.4

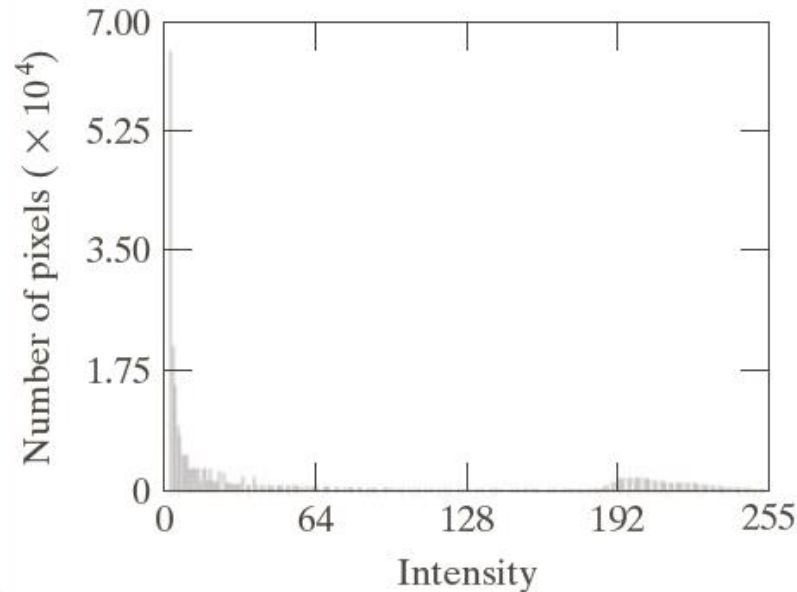
Mappings of all the values of s_k into corresponding values of z_q .

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(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

Histogram Processing : Histogram Matching

- The image of Fig. 2.23 is dominated by **large, dark areas**, resulting into histogram characterized by a large concentration of pixel in the dark end of the gray scale.



a b

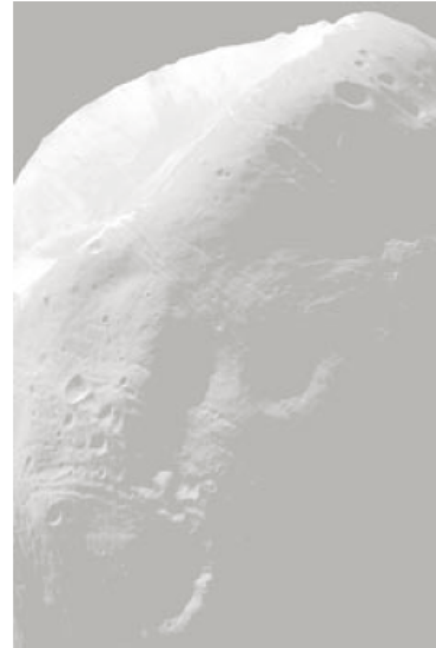
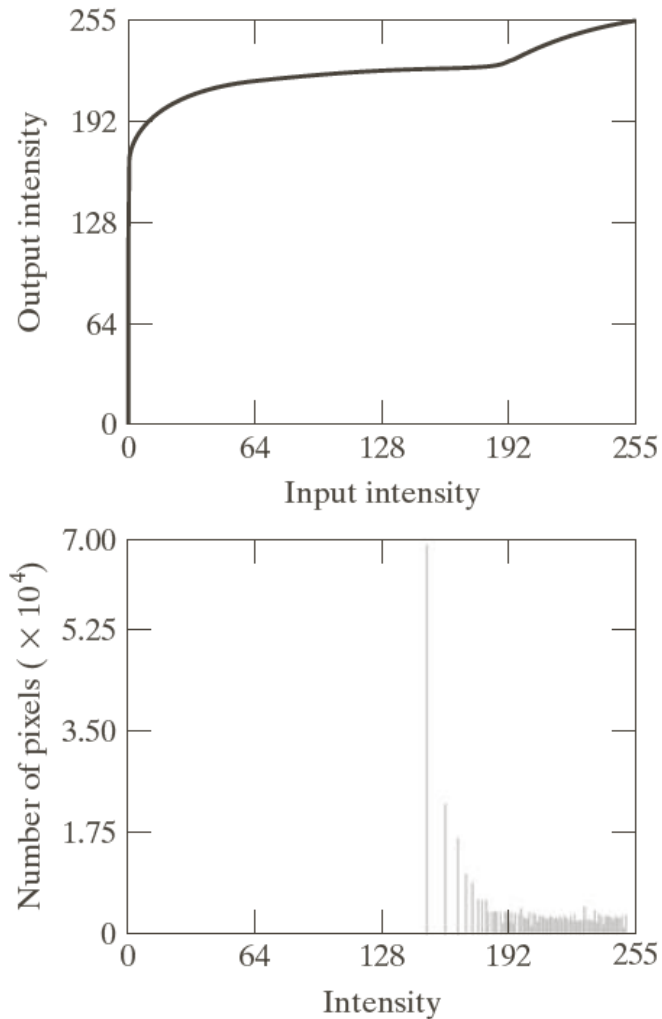
FIGURE 3.23

(a) Image of the Mars moon Phobos taken by NASA's *Mars Global Surveyor*.
(b) Histogram.
(Original image courtesy of NASA.)

(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

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Histogram Processing : Histogram Matching



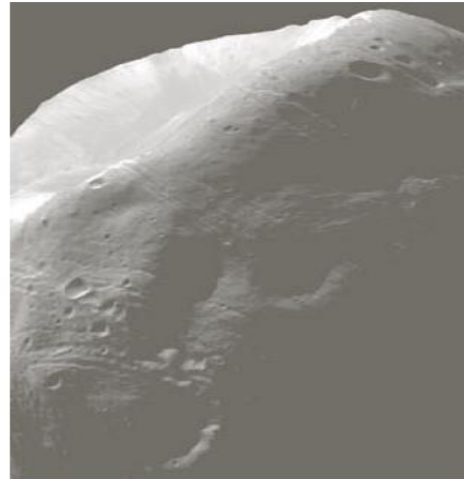
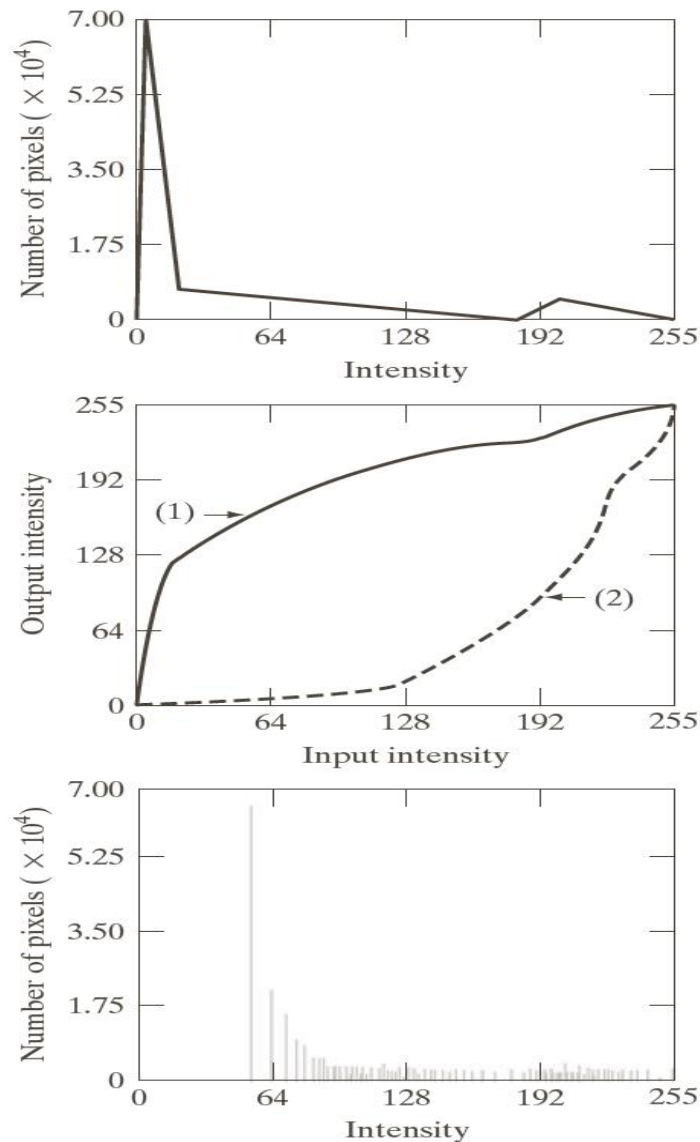
a b
c

FIGURE 3.24
(a) Transformation function for histogram equalization.
(b) Histogram-equalized image (note the washed-out appearance).
(c) Histogram of (b).

(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

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Histogram Processing : Histogram Matching



a c
b
d

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FIGURE 3.25
(a) Specified
histogram.
(b) Transformations.
(c) Enhanced image
using mappings
from curve (2).
(d) Histogram of (c).

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